E-Commerce Business Analysis

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AGENDA

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01

Background

The Evolution of E-Commerce

1.0 Era: Going Online	Traditional retail shifted online, supported by the development of search engines, digital payments, and logistics.
2.0 Era: Platform-Based Commerce	Platforms like Amazon and Alibaba expanded rapidly, while mobile shopping became mainstream.
3.0 Era: Social & Data-Driven Commerce	E-commerce shifted from active product searches to algorithm-driven recommendations and content-based product discovery

Data Science in E-Commerce

Understanding Consumer Behavior

Analyze browsing, purchasing and transaction data to understand customer behavior and preferences.

Customer Segmentation

Group customers based on shared characteristics and behaviors to develop targeted marketing strategies.

Customer Value Analysis

Identify high-value customers and optimize the allocation of marketing resources.

Data-Driven Decision Making

Support targeted marketing, customer retention, personalized recommendations, and strategic business decisions.

02

Data & Methodology

Data Sources

Customers	• Customer IDs, registration dates, and customer information
Orders	• Order amounts, payment status, fulfillment status, and refunds
Order Items	• Products purchased, quantities, and associated order IDs
Products	• Product categories, SKUs, and product details
Traffic	• Website sessions, product views, add-to-cart events, and checkouts
Transactions	• Payments, refunds, and transaction success or failure status

Methodology

Data Cleaning & Integration	• Identify missing values, duplicates, and outliers. • Validate ID consistency across tables. • Merge order, customer, product, and transaction data.
KPI Analysis	• Net Revenue, Average Order Value (AOV), Refund Rate, Cancellation Rate, Conversion Rate, Customer Lifetime Value (CLV)
Customer Segmentation	• Construct RFM features: Recency, Frequency, and Monetary Value. • Standardize numerical features. • Apply K-Means clustering. • Identify four distinct customer segments.
Further Business Analysis	• Product Performance Analysis • Website Conversion Funnel Analysis • Customer Value & Retention Analysis

03

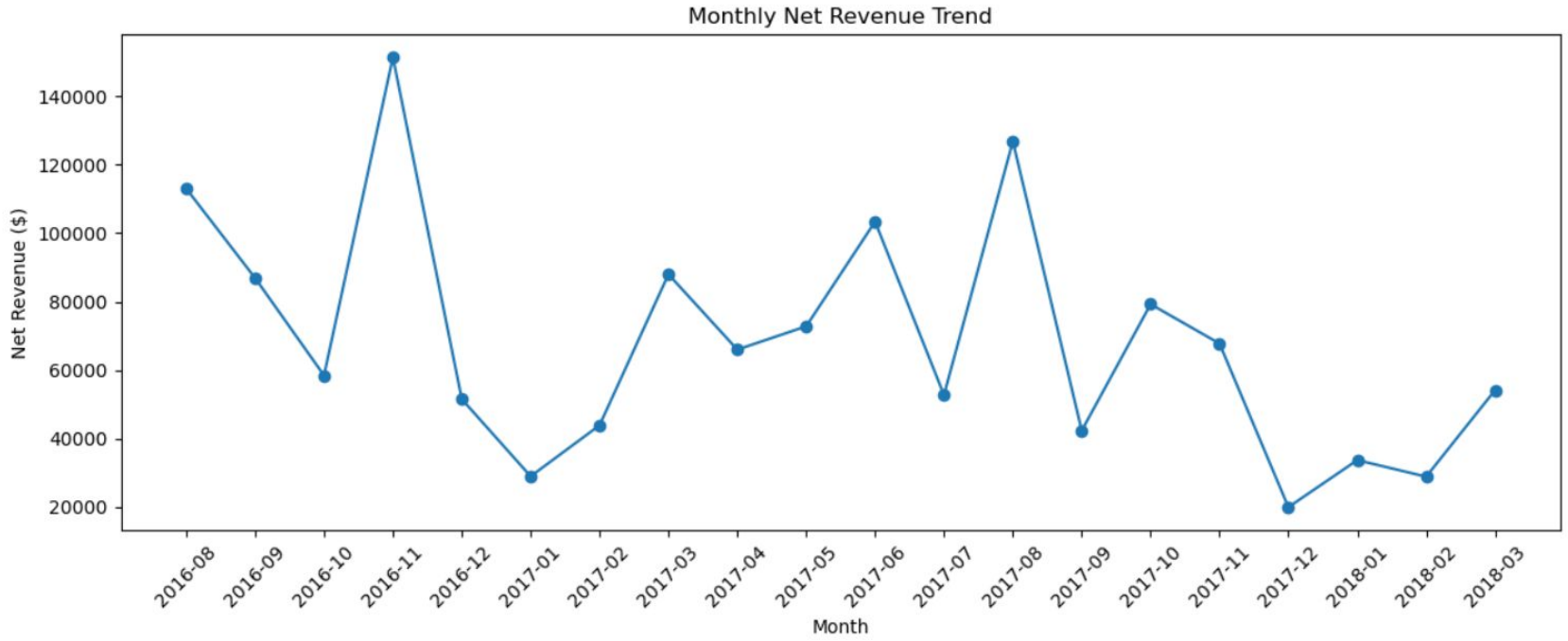
Overall Business Performance

KPI Dashboard

[63]:

	KPI	Value
0	Total Orders	21,358
1	Valid Sales Orders	20,948
2	Valid Order Rate	98.08%
3	Gross Successful Payments	$1,579,872.23
4	Successful Refund Amount	$210,601.78
5	Net Revenue	$1,369,270.45
6	Average Order Value	$65.37
7	Cancellation Rate	1.92%
8	Refund Order Rate	14.91%
9	Refund Amount Rate	13.33%
10	Customers with Valid Orders	14,758
11	Repeat Customers	3,146
12	Repeat Customer Rate	21.32%
13	Average Orders per Customer	1.42

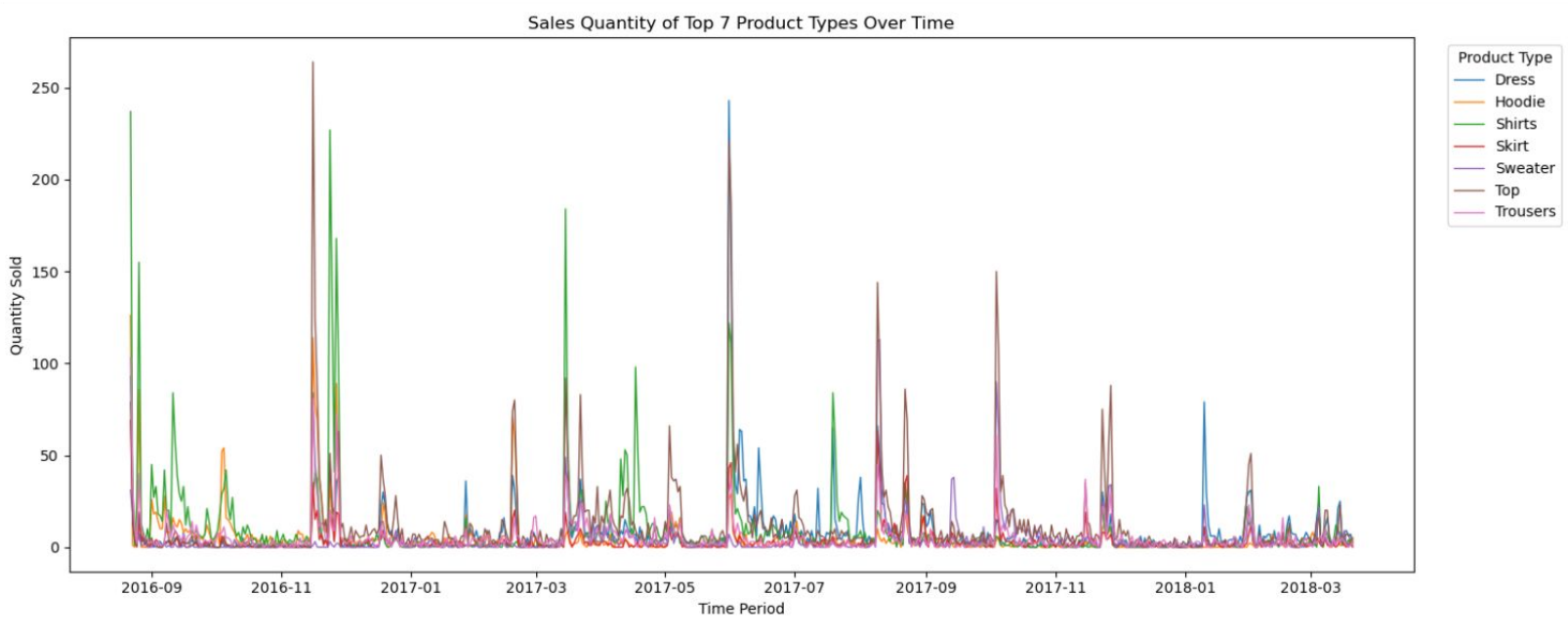
Monthly Net Revenue



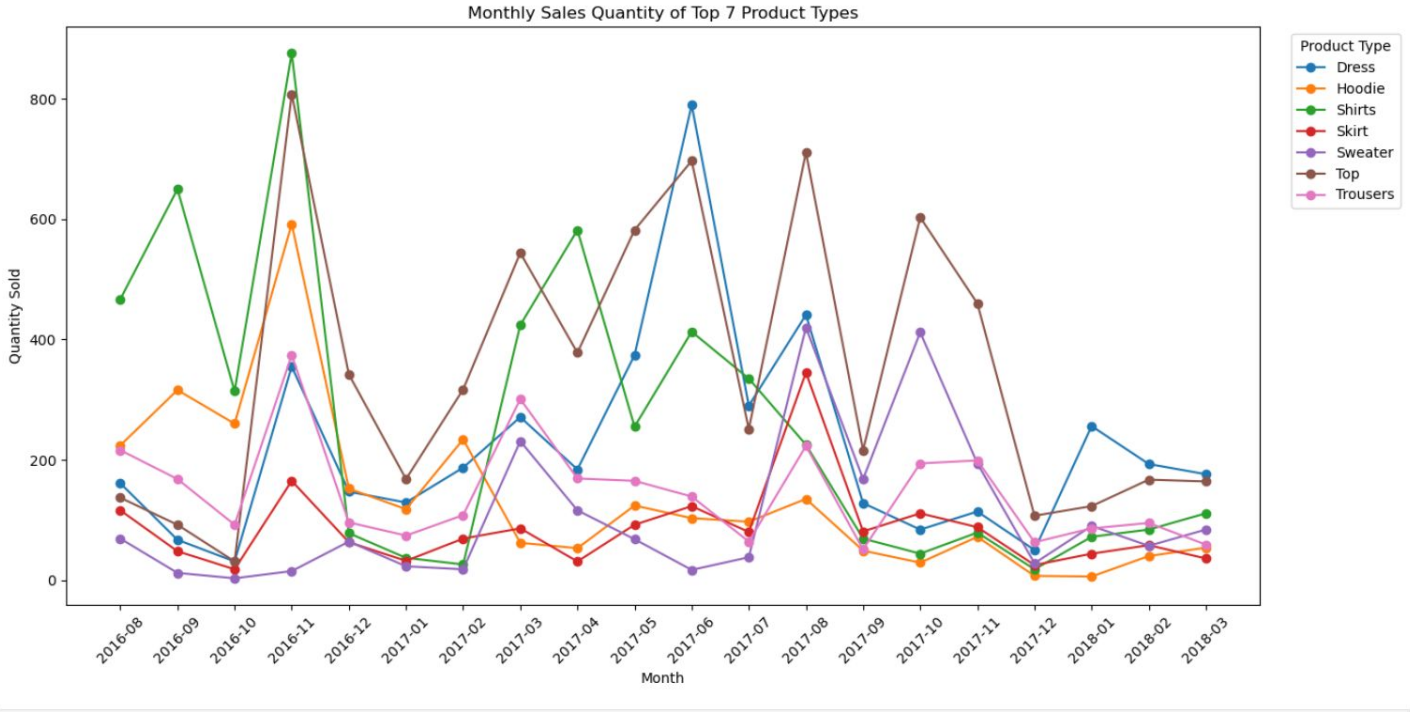
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Product Performance

Top Product Types



Monthly Product Trends

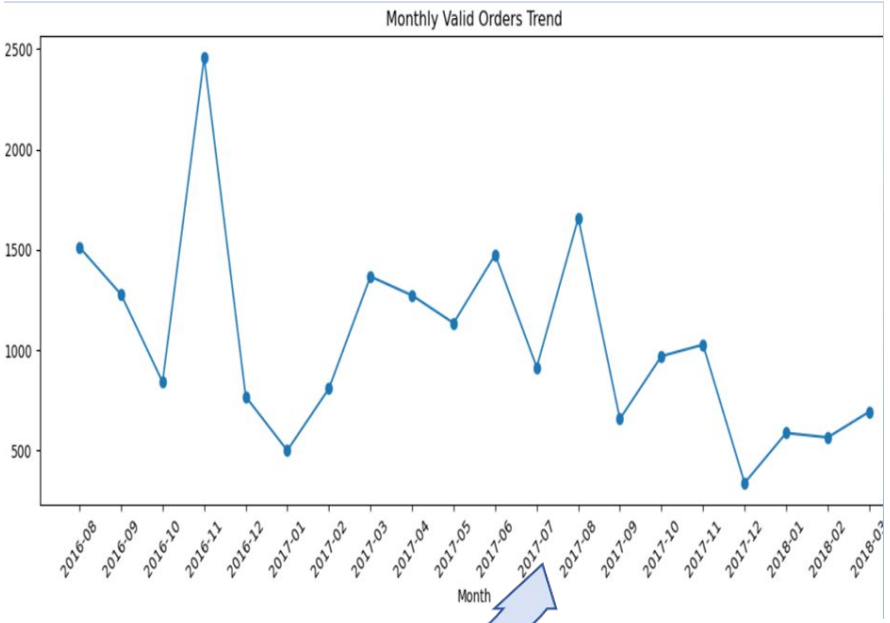


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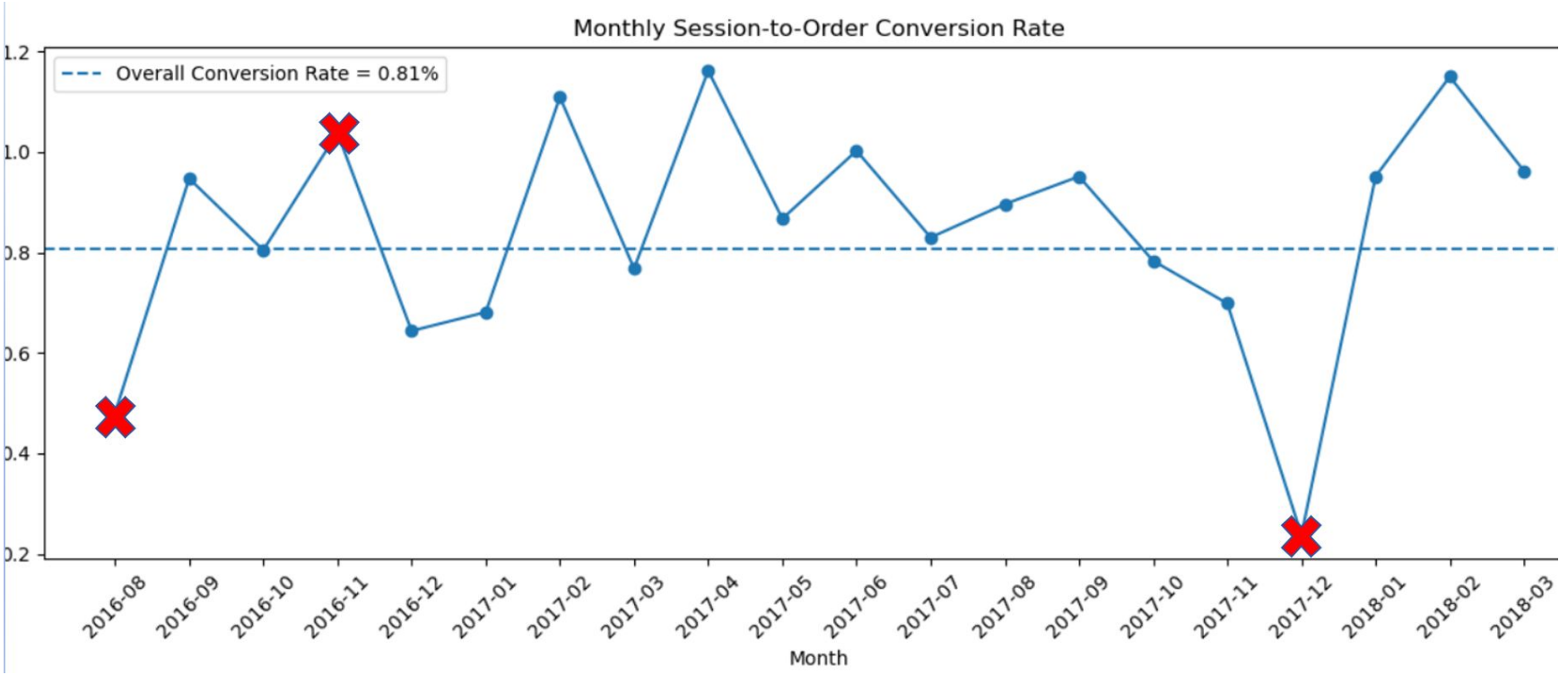
Traffic & Conversion Performance

Monthly Sessions & Orders

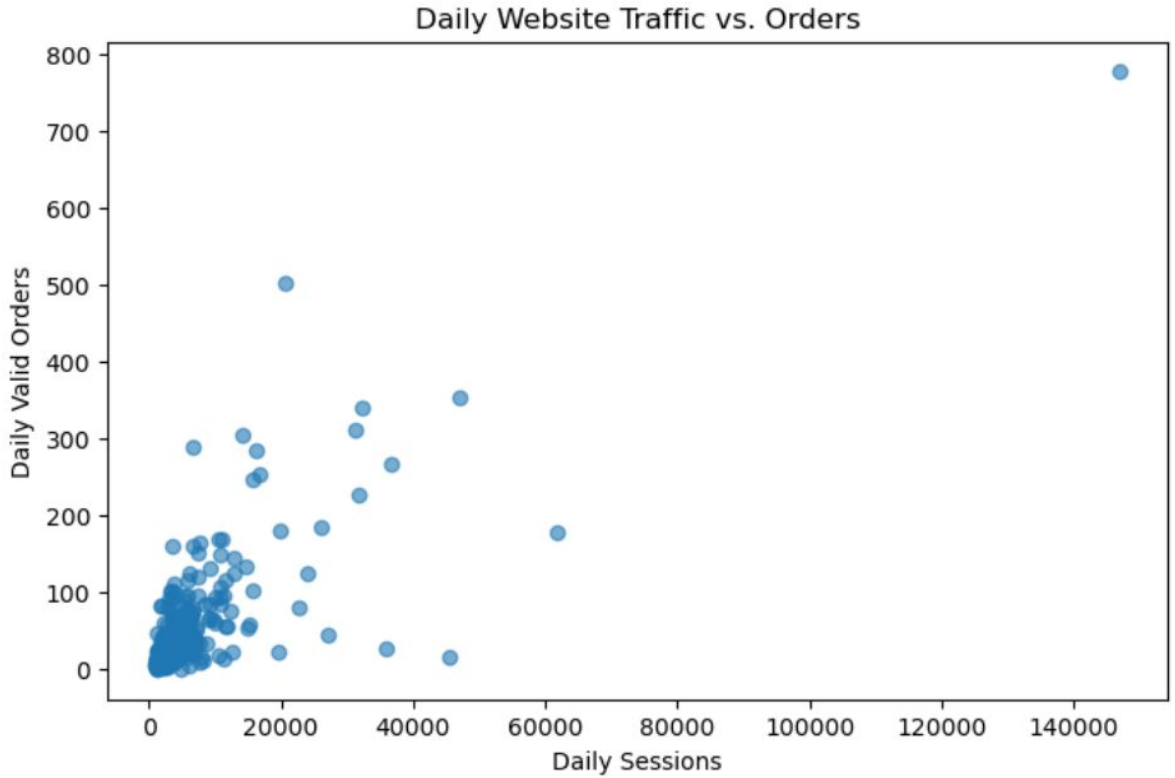




Monthly Conversion Rate

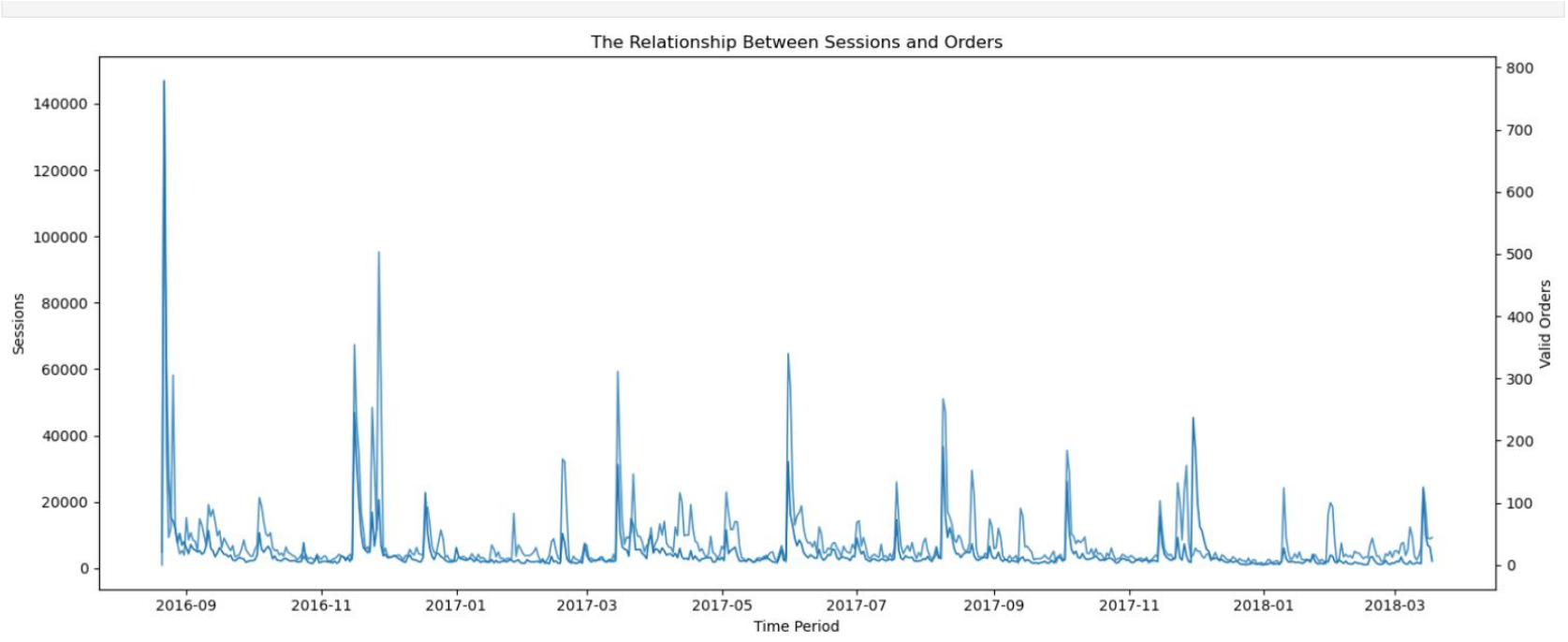


Sessions vs Orders correlation

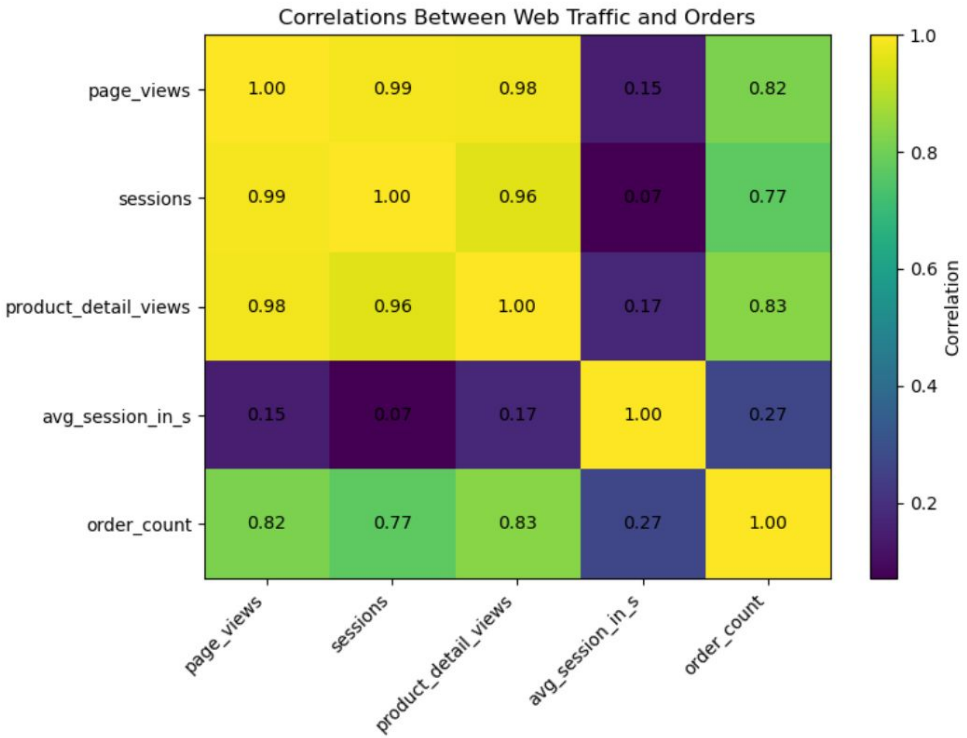


Correlation between Sessions and Valid Orders: 0.770

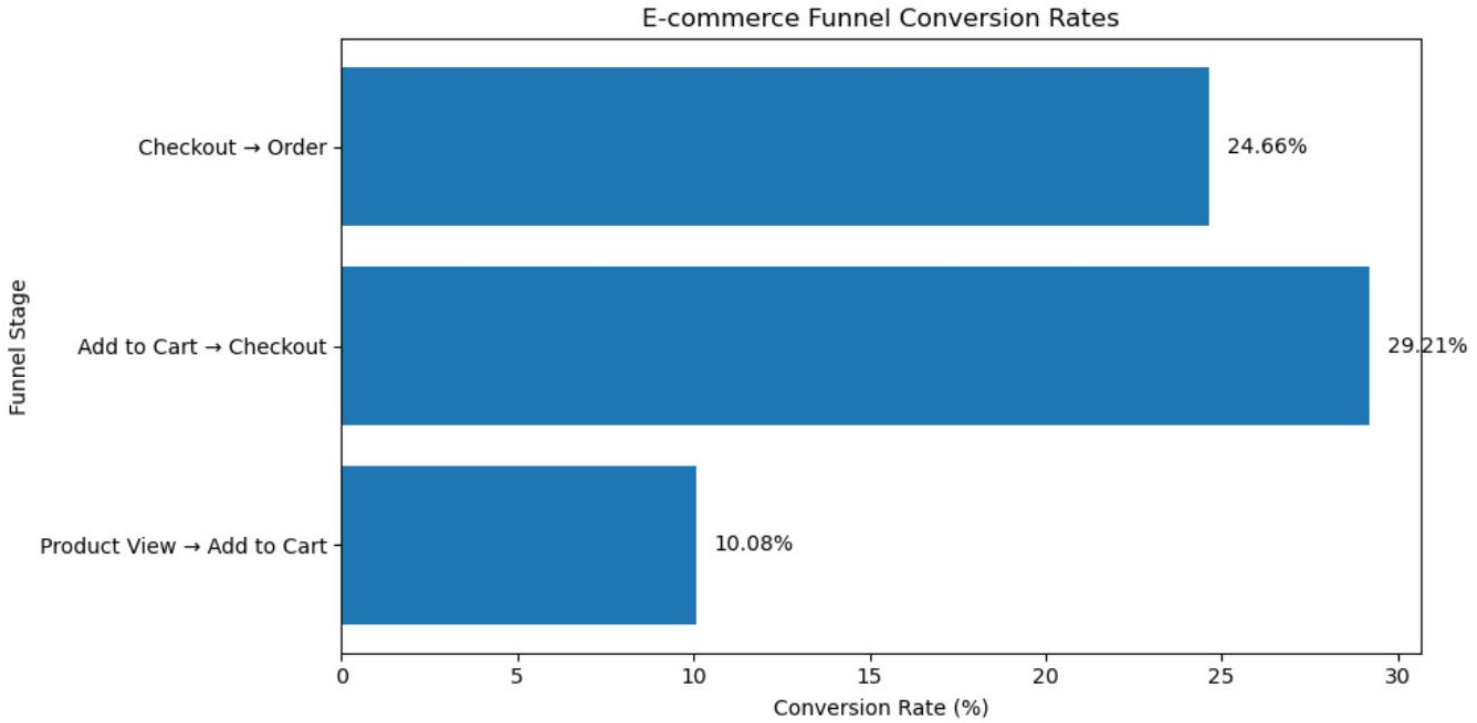
Sessions vs Orders correlation



Sessions vs Orders correlation



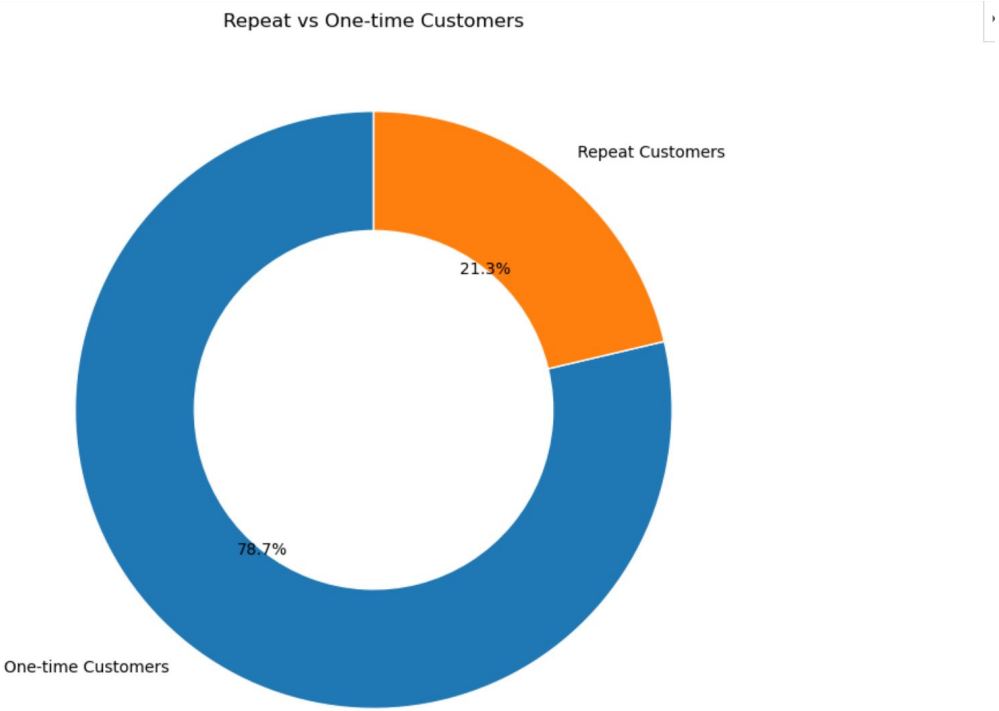
Funnel Conversion



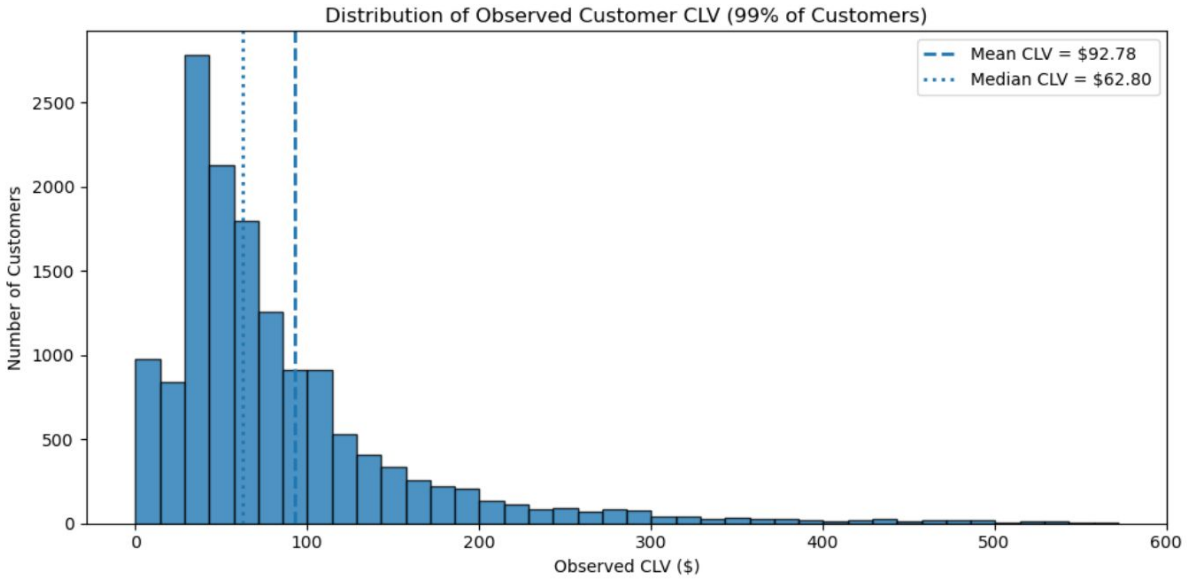
06

Customer Retention & Value

Repeat vs One-Time



Observed CLV Distribution



07

Customer Segmentation with RFM & K-Means

Segmentation Workflow

RFM

Recency
Frequency
Monetary

→

Log Transformation & Standardization

Reduce skewness and the impact of extreme values, and standardize the three RFM features to a comparable scale

→

Optimal K Selection

Apply K-Means clustering with the selected number of clusters **(K = 4)**

→

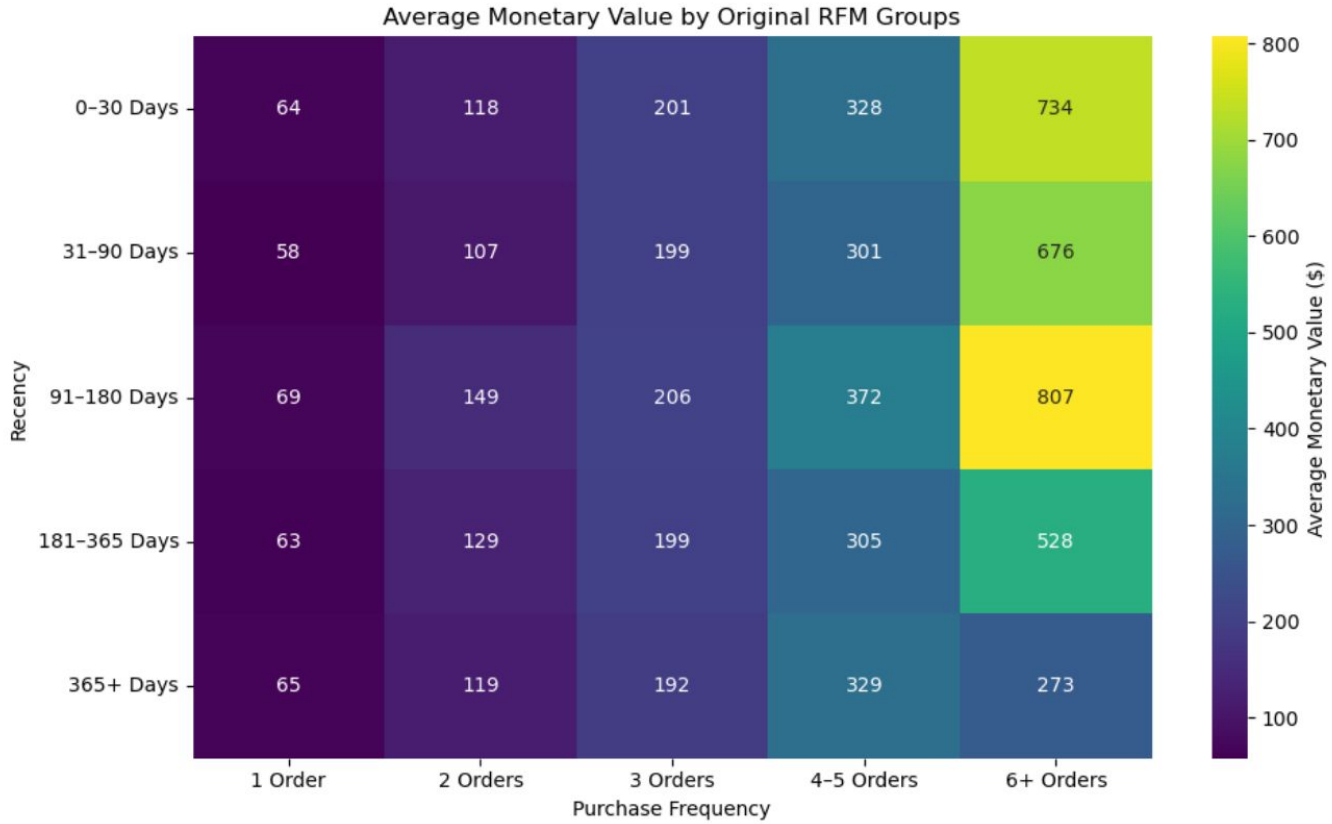
Business Labeling

Interpret each cluster's RFM characteristics and assign meaningful customer segment labels

Understanding RFM Analysis

R— Recency	How recently a customer purchased
F — Frequency	How often a customer purchased
M — Monetary	How much value a customer contributed

RFM Heatmap

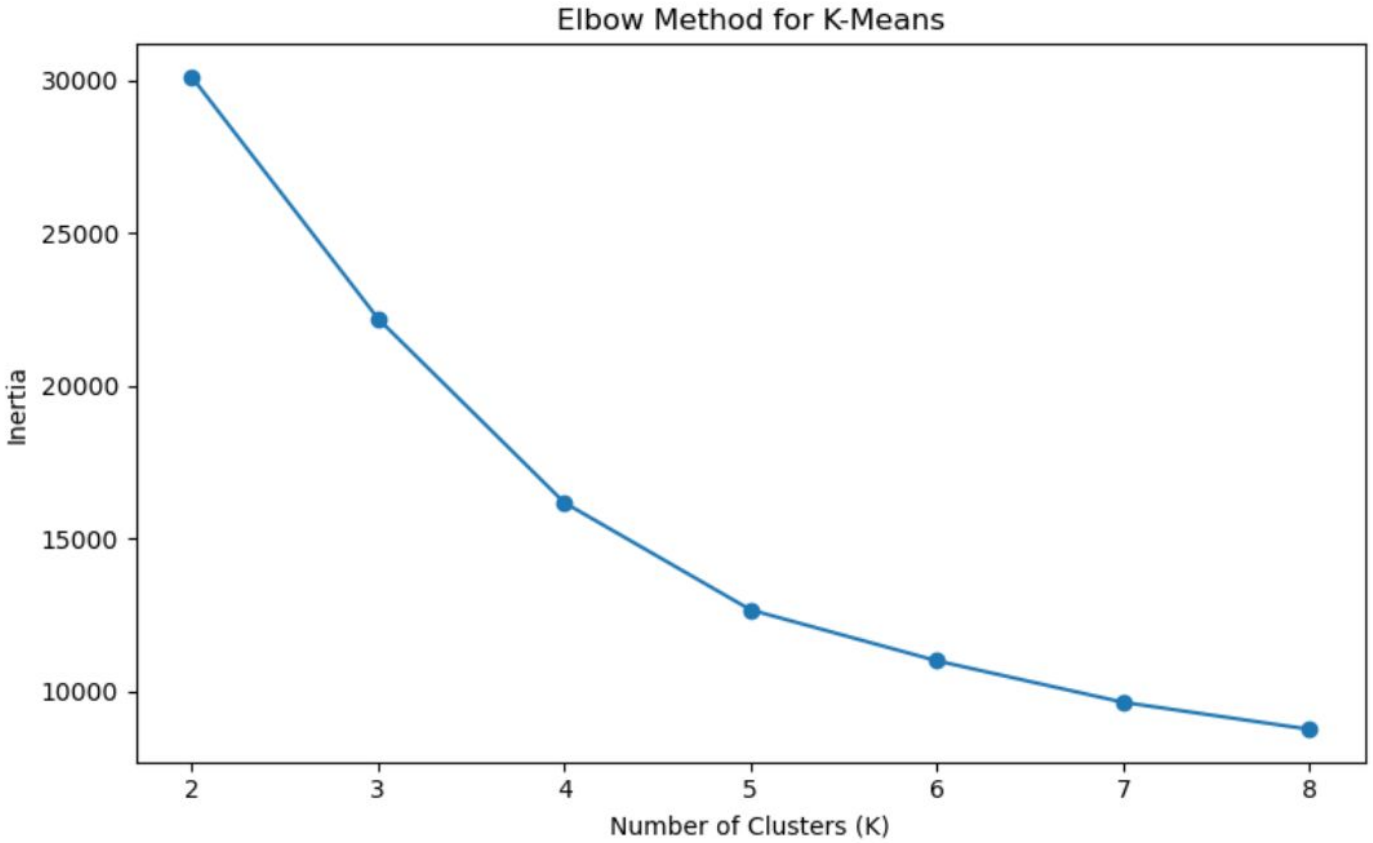

Average Monetary Value by Original RFM Groups
Recency
0–30 Days
31–90 Days
91–180 Days
181–365 Days
365+ Days
64
118
201
328
734
58
107
199
301
676
69
149
206
372
807
63
129
199
305
528
65
119
192
329
273
1 Order
2 Orders
3 Orders
4–5 Orders
6+ Orders
Purchase Frequency
Average Monetary Value ($)
800
700
600
500
400
300
200
100

Transformation & Standardization

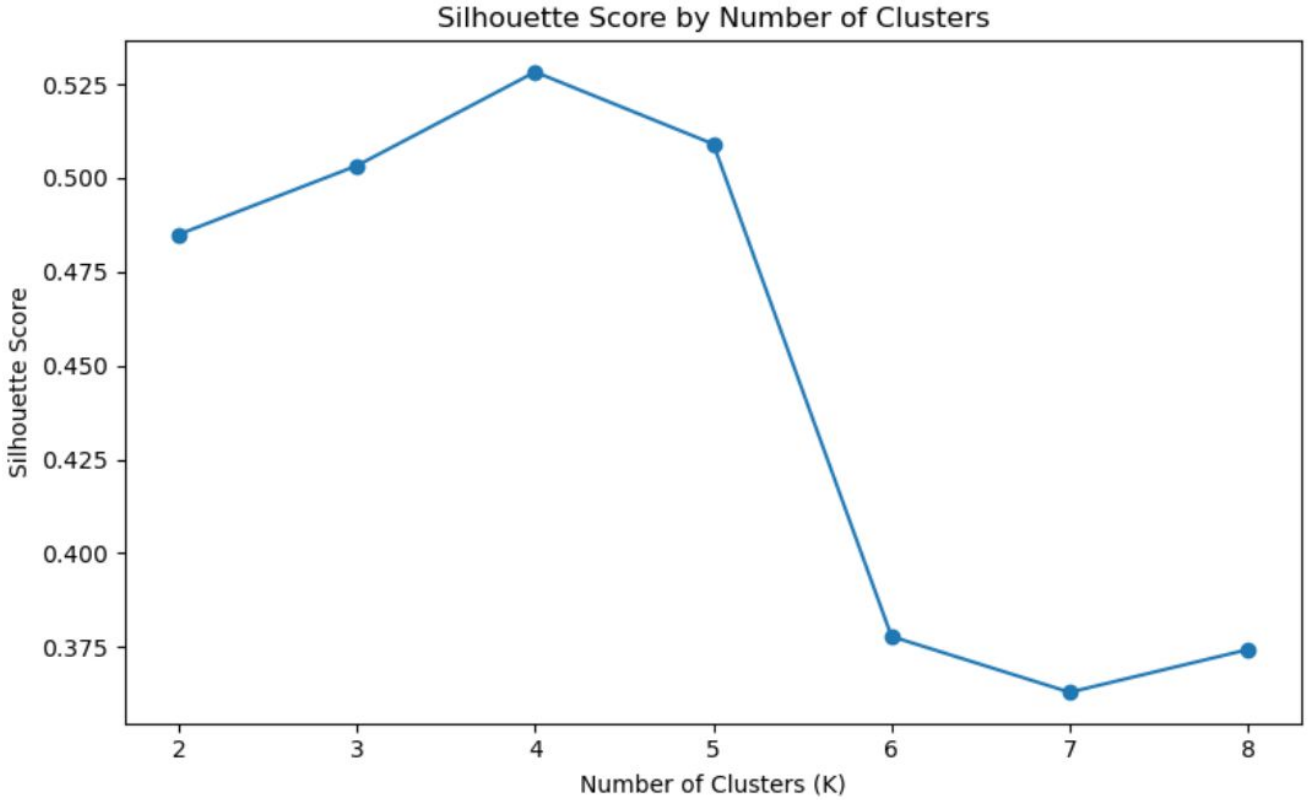
Why?	• RFM variables have different scales • Data is skewed and affected by extreme values
What we did	• Log Transformation → reduce skewness • Standardization → put R, F, M on the same scale

Better input for K-Means clustering

Elbow

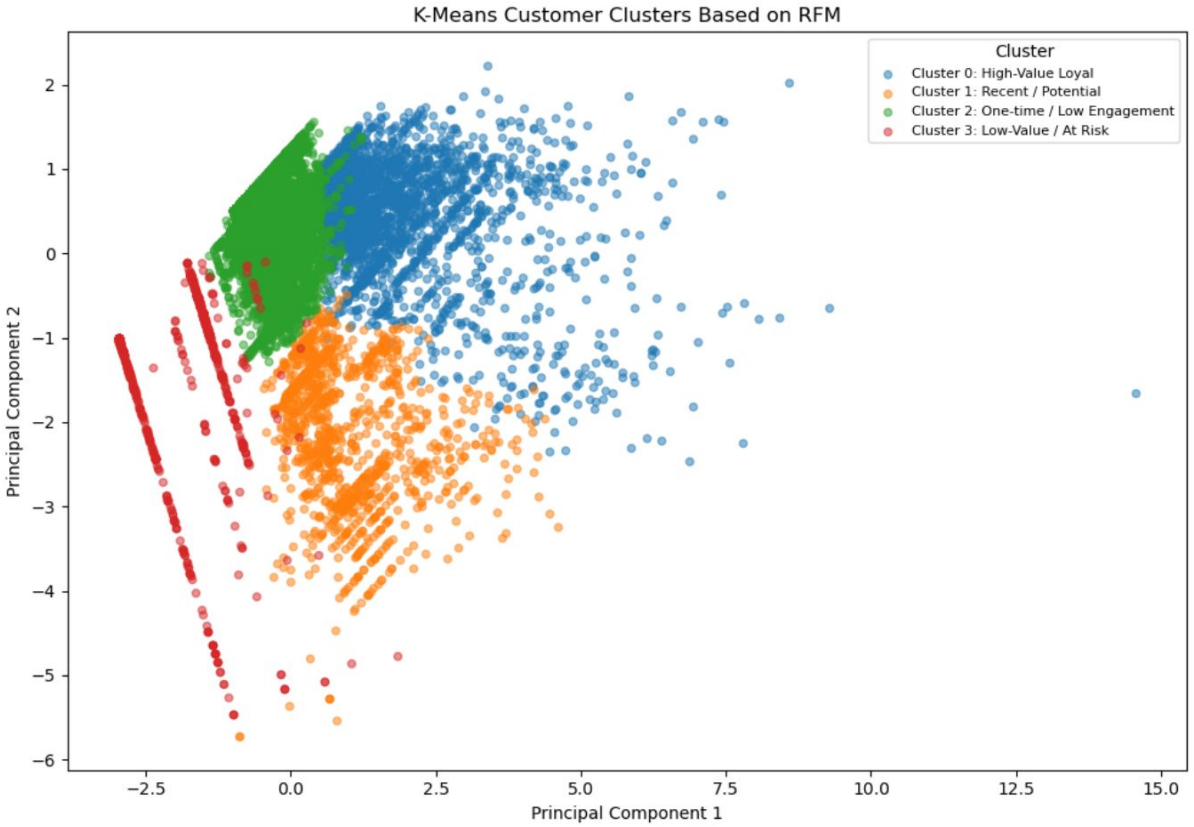

Elbow Method for K-Means
Inertia
30000
25000
20000
15000
10000
2
3
4
5
6
7
8
Number of Clusters (K)

Silhouette


Silhouette Score by Number of Clusters
Silhouette Score
0.525
0.500
0.475
0.450
0.425
0.400
0.375
2
3
4
5
6
7
8
Number of Clusters (K)

PCA

Explained variance by 2 PCs: 82.76%

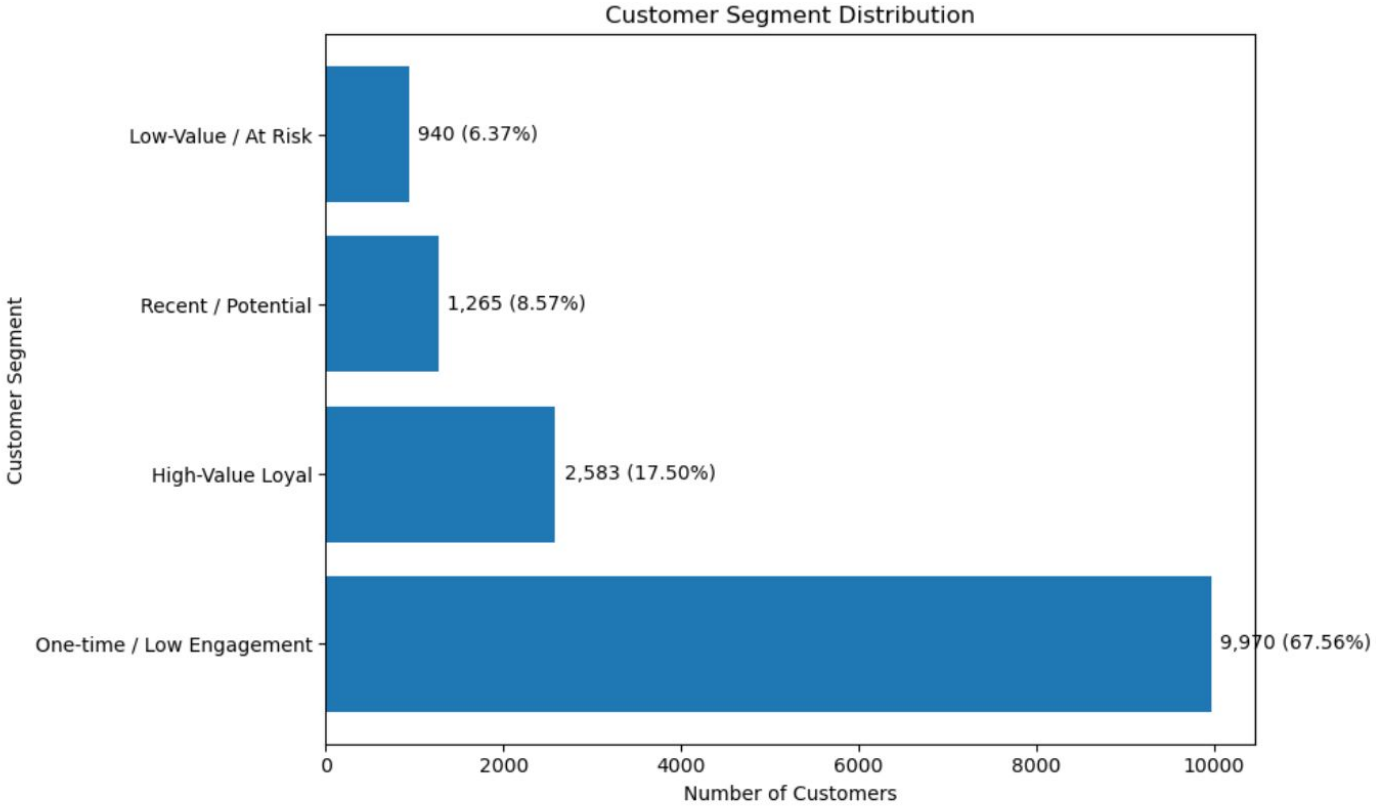


Cluster Profiles

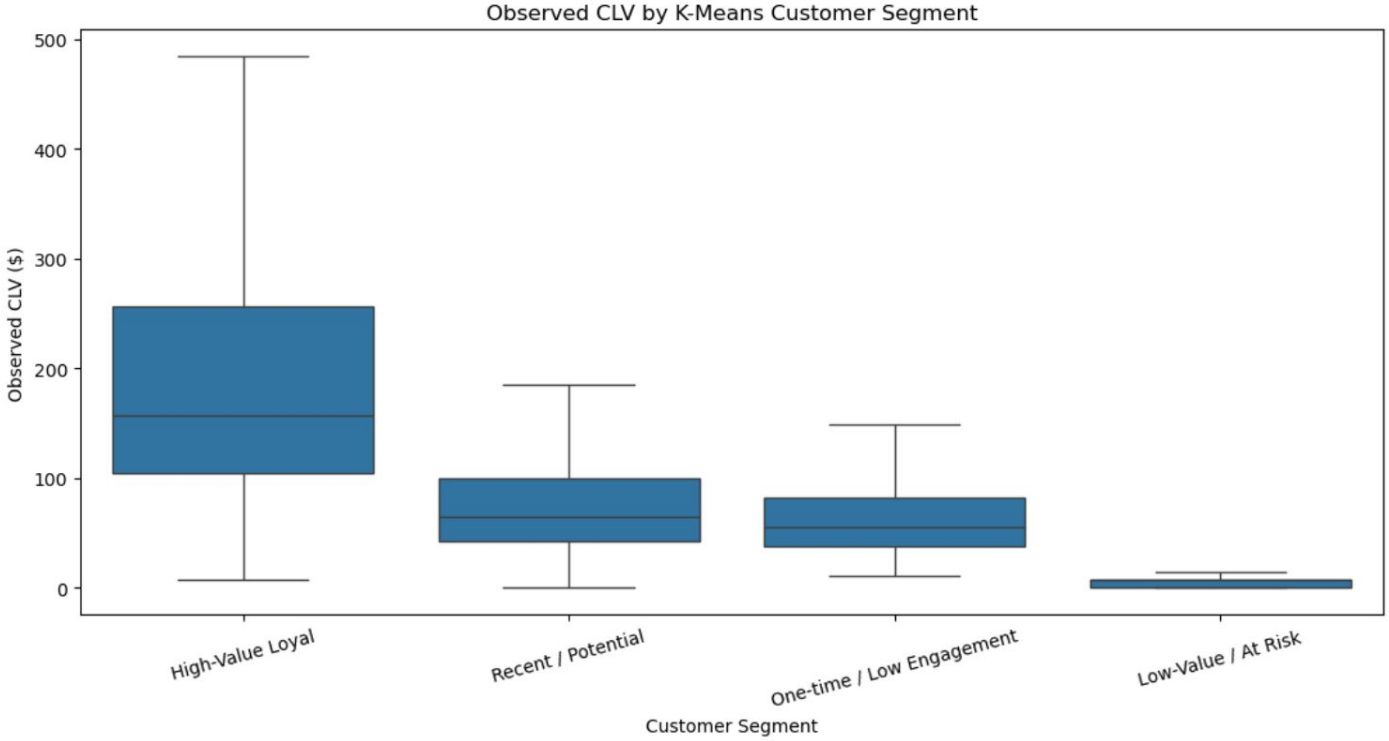
	cluster	customer_count	avg_recency	median_recency	avg_frequency	median_frequency	avg_monetary	median_monetary	customer_percentage
0	0	2583	254.39	233.0	3.14	2.0	224.21	157.00	17.50
1	1	1265	31.30	26.0	1.28	1.0	81.74	65.00	8.57
2	2	9970	369.70	365.0	1.02	1.0	68.54	55.49	67.56
3	3	940	301.08	295.0	1.09	1.0	3.63	7.00	6.37

	customer_segment	customer_count	customer_percentage
0	One-time / Low Engagement	9970	67.56
1	High-Value Loyal	2583	17.50
2	Recent / Potential	1265	8.57
3	Low-Value / At Risk	940	6.37

Customer Segment Distribution



Observed CLV by K-Means Segmentation



08

Recommendations & Conclusion

Conclusion

- Traffic is positively related to orders, but overall conversion remains low.
- Most customers are one-time or low-engagement customers.
- Repeat customers account for only 21.32% of the customer base.
- High-value customers are a smaller group, but they contribute significantly higher CLV.

Recommendations

Increase Repeat Purchases

Finding: Repeat customer rate = 21.32%

Action: Post-purchase campaigns, incentives and personalized recommendations.

Reduce Revenue Leakage

Finding: Refund amount rate = 13.33%

Action: Identify high-refund products and investigate refund drivers.

Improve Funnel Conversion

Finding: Overall session-to-order conversion = 0.81%

Action: Improve product pages, cart experience and checkout process.

Apply Segment-Based Marketing

Finding: 67.56% = One-time / Low Engagement

Action: Different marketing strategies for different customer values.

Questions?

THANK YOU

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